Isogeny-loop GLV, round 2: open leads, an erratum, and 4-dimensional GLV-GLS

2026-10-11 · study directory `research/isogeny_loop_glv_20261011/` · follows `RESEARCH_ISOGENY_LOOP_GLV.md` (PR #1644) · status words: **measured** (native run recorded it), **derived** (follows by the stated argument), **proposed** (untested)

1. Corrections to round 1

Language rule. Round 1's sweeper is Python (cypari2), in breach of AGENTS.md "Implementation language: no Python" (in force since 2026-10-03). Its outputs stay as frozen evidence with Python provenance; every new number here comes from native Rust (`examples/isogeny_loop_degree_bound.rs`, `examples/isogeny_loop_glv_gls_lattice.rs`). Porting the sweeper needs a native subexponential class-group algorithm the crate lacks.

Erratum (derived). Round 1 charged both directions of a characteristic-2 two-step 2 M. Only the Frobenius F is that cheap; the conjugate step is the Verschiebung $V = [2] \circ F^{-1}$. Original statement (§4.3): the folklore split $\lambda = t - 1$ via V^n has ratio 0.49, "not established". Correction: $V^n = [2^n] \circ \pi^{-1}$ is n doublings plus square roots, spent to save $n/2$, so the ratio is $\geq 2(c_{\mathrm{dbl}} + 2)/c_{\mathrm{dbl}} = 2.5$: a provable loss. Every other round-1 conclusion stands a fortiori, because raising a weight only raises loop costs.

2. A class-group-free lower bound (derived; native table)

A loop of prime-degree steps is an endomorphism α with $N(\alpha) = \prod l_i$ and cost $\sum c(l_i) \geq \kappa \log_2 N(\alpha)$, $\kappa = \min_l c(l)/\log_2 l$. A non-scalar $\alpha = \left(A + B\sqrt{D}\right)/2$ has $B \neq 0$, hence $N(\alpha) \geq d_{\min}(D) = \lfloor (|D| + 1)/4 \rfloor$, and $d_{\min}$ only grows in sub-orders. So every loop endomorphism costs at least $\kappa \log_2 d_{\min}(D_K)$ — no class group needed. κ = 14.196 (paper's projective $7.5l$), 5.047 (affine $2l + 2$), 3.786 (optimistic x-only, 3-isogeny eval $4M + 2S$).

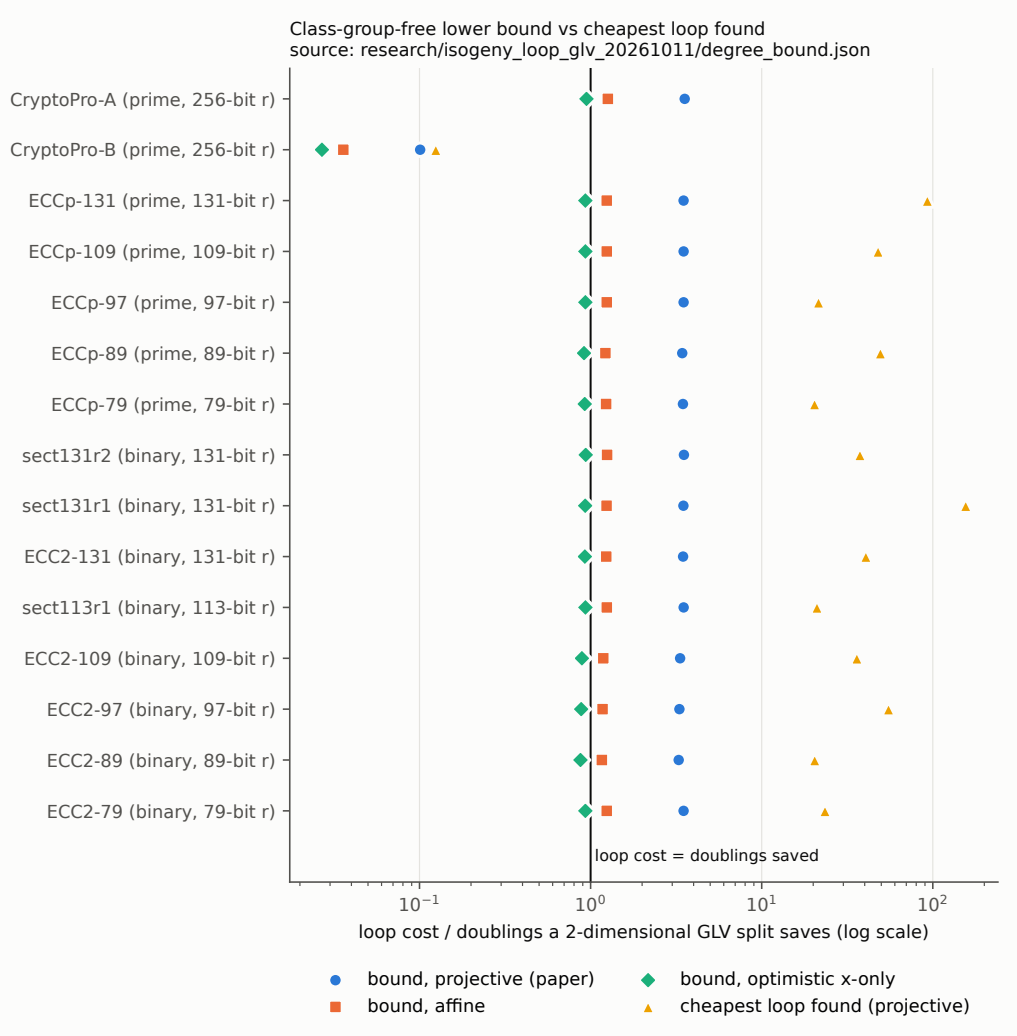


Figure 1: Bound (three per-step models) and the cheapest loop round 1 found, as a fraction of the doublings a 2-dimensional split saves ($8 \cdot \lceil \log_2 r/2 \rceil$ M). Run `run-20261011T2258Z-degree-bound`; every $t^2 - 4q = f^2 D_K$ re-verified natively. Binary "found" values are round 1's under-priced costs, i.e. lower bounds on the corrected costs.

For a random curve $|D_K| \approx q \approx r$, so the ratio is $\approx 2\kappa/c_{\mathrm{dbl}}$: 3.27–3.55 (projective) and 1.16–1.26 (affine) on every random prime and binary challenge curve, **including CryptoPro-A**, whose class group round 1 could not compute. A profitable loop on a random curve would need a step formula below $c_{\mathrm{dbl}}/2 = 4$ M per bit of degree; the optimistic x-only model (0.87–0.95) is not excluded by this bound, only by round 1′s searches (cheapest loops 20–156× the saving, projective). Scope: separable steps; binary loops using F are covered by round 1′s exhaustive search only.

3. Rho on Koblitz curves (derived; heuristic labelled)

An extra rho equivalence needs $\alpha \in O_K$ whose eigenvalue has finite order k outside $\langle -1, \lambda_\tau \rangle$. As $O_K^* = \{\pm 1\}$, $r \le N(\alpha^k - 1) \le \left(N(\alpha)^{k/2} + 1\right)^2$, so $N(\alpha) \gtrsim r^{1/k}$; canonicalising an orbit costs $k - 1$ evaluations, about $\kappa \log_2 r \approx 1800$ M per rho step (projective) to gain $\sqrt{k}$, needing $k \gtrsim 5 \cdot 10^4$ against an 8 M addition. Heuristically the minimum norm over the k cosets is $\approx r/k$, which closes the lead for every k. Open only for an unknown orbit invariant (as the normal-basis rotation is for τ).

4. 4-dimensional GLV-GLS with a moderate-D loop (measured lattice; derived model)

On the quadratic twist $E'/\mathbb{F}_{p^2}$ of a CM curve $E/\mathbb{F}_p$ ($4p = t^2 + |D|\, v^2$), ψ has eigenvalue $\mu = (p-1)/t$ ($\mu^2 = -1$), and the loop endomorphism $\alpha' = \varphi\alpha\varphi^{-1}$ has $\lambda = (A + Bs)/2$, $s = (2\mu - t)/v$, both derived from the ring map $\pi \mapsto \mu$ and asserted per instance. Native LLL (`lll_reduce_hp`) and Babai rounding over 10,000 seeded scalars per lattice, p of 127 bits.

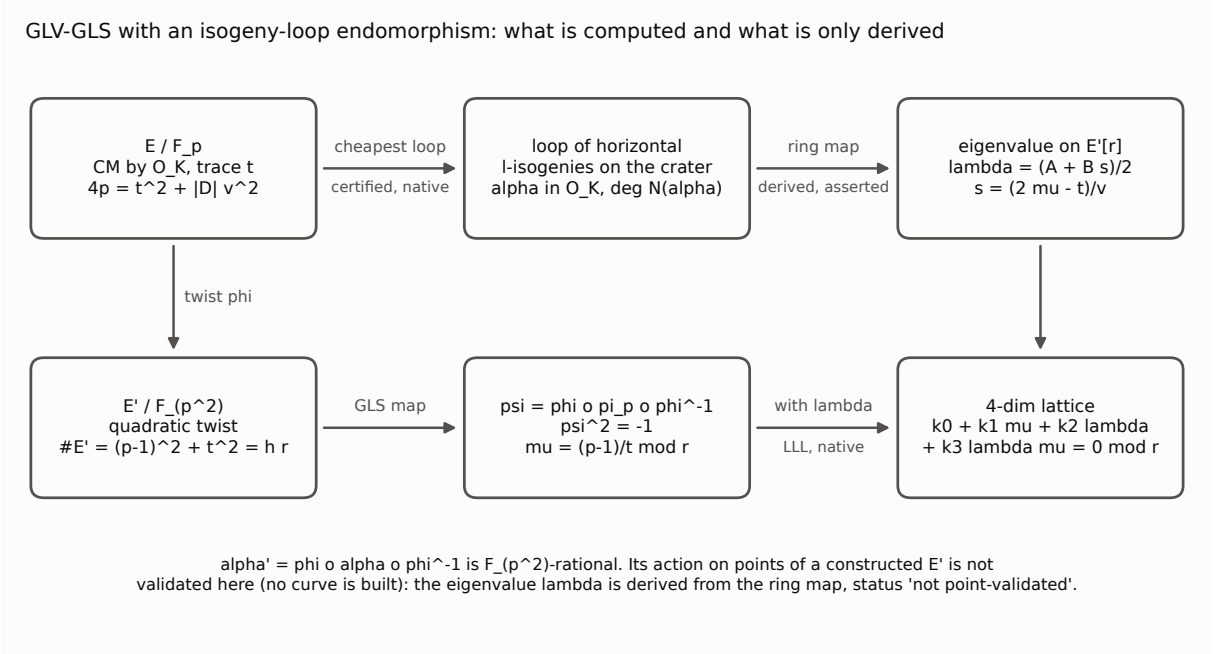


Figure 2: What is computed natively, what is derived, and what is not validated (no curve is constructed; λ is not checked on points).

D	cheapest loop	cost M	h	$\log_2 r$	GLS-2 max/mean	GLV-GLS-4 max/mean	model ratio
−3	aut. ω (control)	1	1	254	126 / 125.5	63 / 62.6	1.357
−4	aut. i	1	—	—	no instance (twist order splits)	—	—
−7	2	15	4	251	125 / 124.1	63 / 61.9	1.340
−8	2	15	8	250	125 / 123.5	63 / 61.7	1.339
−11	3	22.5	20	249	124 / 123.0	62 / 61.4	1.332
−23	2·3	37.5	52	248	124 / 122.8	62–63 / 61.2–61.3	1.316
−71	2·3²	60	20	249	124 / 123.0	62–63 / 61.4	1.292
−431	2²·3³	97.5	20	249	124 / 122.9	63 / 61.7–62.1	1.248 / 1.241
−619	5²·7	127.5	25	250	125 / 123.5	62–63 / 61.6–62.1	1.220 / 1.228
−3019	5³·13	210	5	251	125 / 124.2	63 / 62.0	1.158

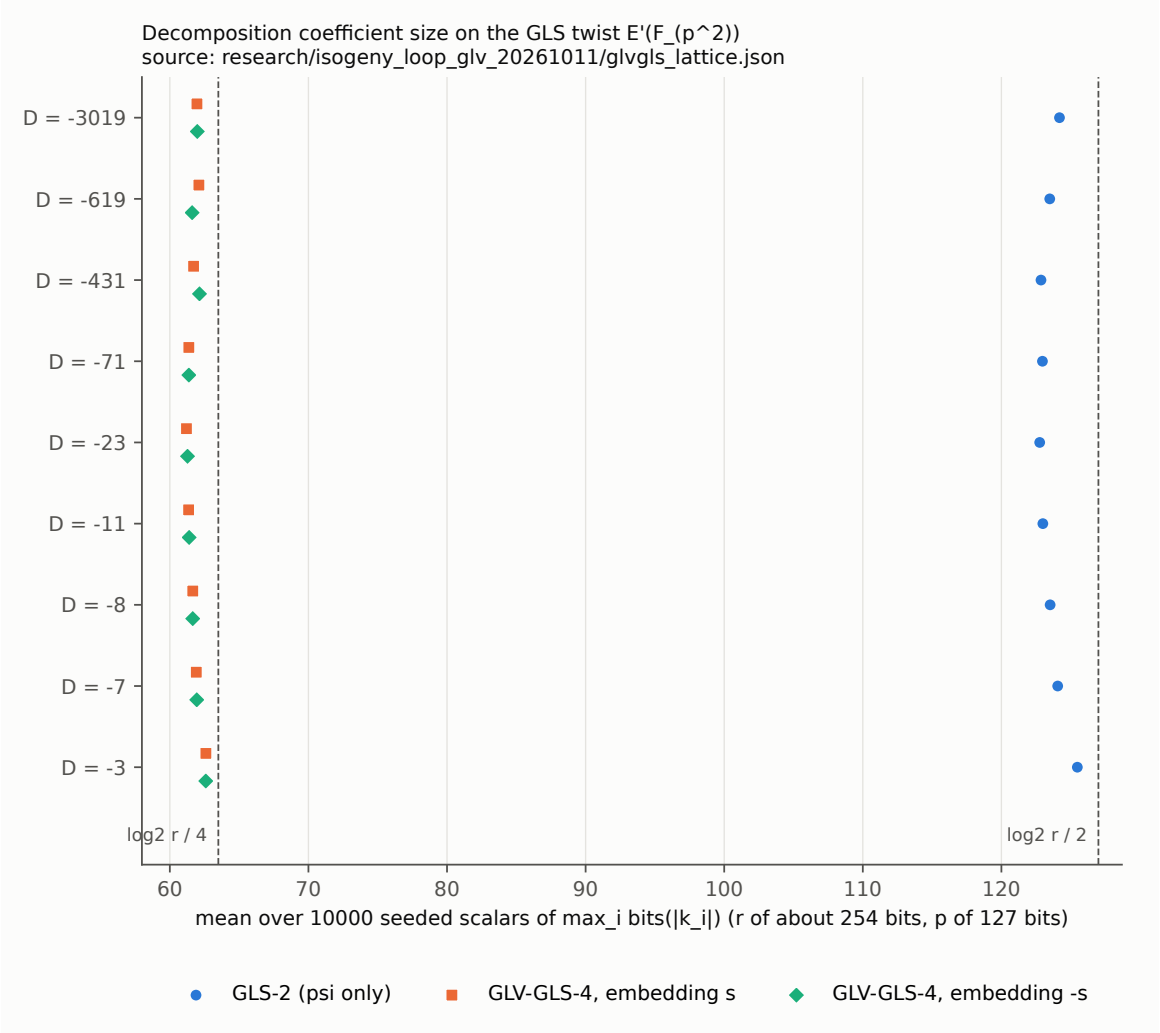


Figure 3: Mean over 10,000 scalars of $\max_i \mathrm{bits}(|k_i|)$; both ring embeddings $(s, -s)$ shown. Run `run-20261011T2258Z-glvgls-lattice`.

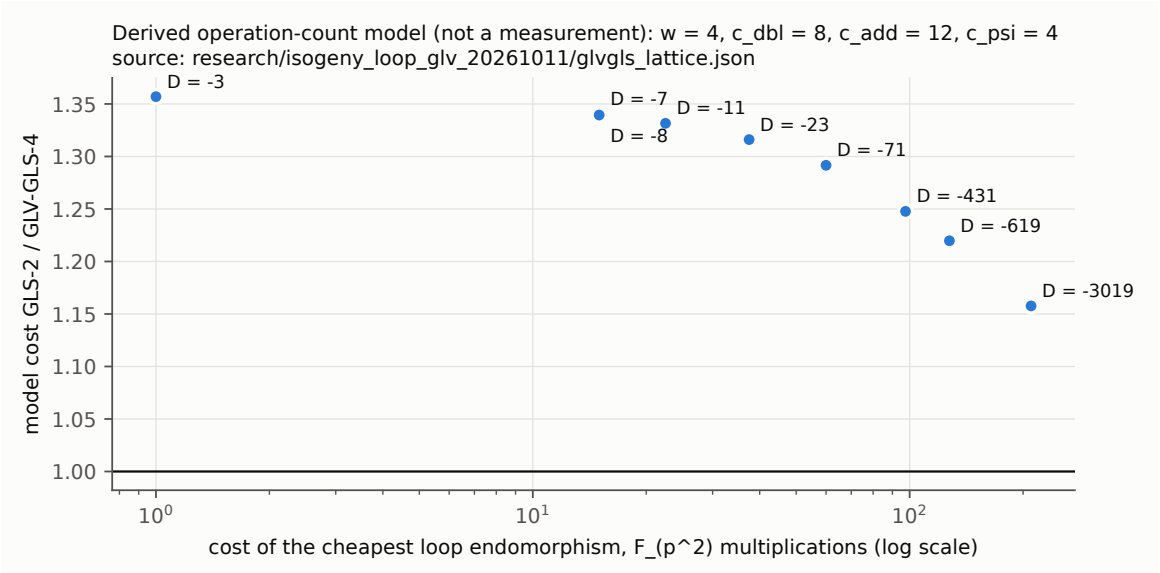


Figure 4: Derived operation-count model (not measured): interleaved $w = 4$ NAF, $c_{\mathrm{dbl}} = 8$, $c_{\mathrm{add}} = 12$, $c_\psi = 4$ $\mathbb{F}_{p^2}$ multiplications, loop charged once.

Measured: cheapest loops are certified exhaustively up to $N \le 2^{C/\kappa}$ and reproduce round 1 and the paper ($D = -619$: 127.5 M vs 128; $D = -3019$: $5^3 \cdot 13$, 210 M, Table 10). The 4-dimensional lattice reaches 62–63 bits ($\log_2 r/4 = 62$–63.5) for every D, as for the control. **Derived:** $D = -4$ has no quadratic-GLS instance, since with $x = t/2$, $(p-1)^2 + t^2 = (x^2 + (1+v)^2)(x^2 + (1-v)^2)$ (a first run stalled 13.5 CPU-minutes there and was stopped; kept as a failed attempt). **Not established:** λ on points of a constructed E', and any timing.

5. Crate issue, and open leads

`lattice::lll_reduce` (f64 Gram–Schmidt) returned the 254-bit GLS-2 lattice unreduced without error (189-bit vectors vs 126 with `lll_reduce_hp`); reported, not fixed here.

lead	**status**	**what settles it**
native sweeper port (130-bit class groups)	required by AGENTS.md before more sweeps; not attempted	native subexponential class-group algorithm
point-level validation of λ on a GLS curve	open	native Hilbert class polynomials for $h > 1$, twisted loop over $\mathbb{F}_{p^2}$
GLV-GLS-4 ($D = -619$) implementation timing	open	Rust benchmark as in `cryptopro_b_glv_chain_20261005`
binary GLS analogue	proposed	same lattice run, binary cost constants
random curves under x-only costs	bound inconclusive (≈ 0.93×)	native x-only re-run of the searches
extra rho equivalences on Koblitz curves	closed (evaluate-the-orbit model)	a new orbit invariant
folklore split $\lambda = t - 1$ on binary curves	closed: provable loss	—

Sources: Koshelev, Sanso, ePrint 2024/1985 (CIFRIS25 ACTA, doi:10.69091/koine/vol-7-W09); Galbraith, Lin, Scott, ePrint 2008/194; Longa, Sica, ASIACRYPT 2012. Records: `degree_bound.{md,json}`, `glvgls_lattice.{md,json}`, `revision.txt` (base commit db0393b4, rustc 1.97.0, source SHA-256).